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1. 简述

硬件: IMX8MQ

参考芯片手册: IMX8MQ 芯片手册第十章

内核:

kernel/include/linux/spi/spi.h

kernel/Documentation/devicetree/bindings/spi/spi-controller.yaml

kernel/drivers/spi/spi.c

2. 设备树头文件

/*

参考:

1. IMX8MQ 芯片手册第七章-7.1.4 SDMA event mapping(1721)
2. kernel/Documentation/devicetree/bindings/dma/fsl-imx-sdma.txt
3. kernel/arch/arm64/boot/dts/freescale/imx8mq.dtsi
4. kernel/Documentation/devicetree/bindings/spi/fsl-imx-cspi.txt

*/

```
ecspi1: spi@30820000 {
    #address-cells = <1>;
    #size-cells = <0>;
    compatible = "fsl,imx8mq-ecspi", "fsl,imx6ul-ecspi";
    reg = <0x30820000 0x10000>;
    interrupts = <GIC_SPI 31 IRQ_TYPE_LEVEL_HIGH>;
    clocks = <&clk IMX8MQ_CLK_ECSP1_ROOT>,
    <&clk IMX8MQ_CLK_ECSP1_ROOT>;
    clock-names = "ipg", "per";
    // dmas = <&sdma1 0 7 1>, <&sdma1 1 7 2>;
    // dma-names = "rx", "tx";
    status = "disabled";
};
```

```
ecspi2: spi@30830000 {
    #address-cells = <1>;
    #size-cells = <0>;
    compatible = "fsl,imx8mq-ecspi", "fsl,imx6ul-ecspi";
    reg = <0x30830000 0x10000>;
    interrupts = <GIC_SPI 32 IRQ_TYPE_LEVEL_HIGH>;
    clocks = <&clk IMX8MQ_CLK_ECSP2_ROOT>,
    <&clk IMX8MQ_CLK_ECSP2_ROOT>;
    clock-names = "ipg", "per";
```

```

//dmas = <&sdma1 2 7 1>, <&sdma1 3 7 2>;
//dma-names = "rx", "tx";
status = "disabled";
};

```

这里芯片原厂已经提供 SPI 设备树基础配置，还有中断、DMA。

查找 compatible = “fsl,imx8mq-ecspi”, “fsl,imx6ul-ecspi”关键词，在 drivers/spi/spi-imx.c 中可以找到主机驱动。

3. 开发板设备树

/*

参考:

1. kernel/Documentation/devicetree/bindings/spi/spi-controller.yaml

*/

```

&ecspi1 {
    status = "okay";
    pinctrl-names = "default";
    fsl,spi-num-chipselects = <1>; /* 只有一个设备 */
    pinctrl-0 = <&pinctrl_spi1>; /* pinctrl 节点 */
    cs-gpios = <&gpio5 9 0>; /* 片选引脚 */

    mcp2518fd@0 { /* 这里的 0 指的是 SPI 通道 0 */
        compatible = "microchip,mcp2518fd"; /* 用于匹配设备驱动 */
        reg = <0>; /* 这里的 0 指的是 SPI 通道 0 */
        spi-max-frequency = <20000000>; /* 最高频率 */
    };
};

```

```

&ecspi2 {
    status = "okay";
    pinctrl-names = "default";
    fsl,spi-num-chipselects = <1>;
    pinctrl-0 = <&pinctrl_spi2>;
    cs-gpios = <&gpio5 13 0>;

    spidev@0 {
        compatible = "rohm,dh2228fv";
        reg = <0>;
        spi-max-frequency = <25000000>;
    };
};

```

```

&iomuxc {
    pinctrl_spi1: spi1 {

```

```

fsl,pins = <
    MX8MQ_IOMUXC_ECSP1_SCLK_ECSP1_SCLK    0x82
    MX8MQ_IOMUXC_ECSP1_MOSI_ECSP1_MOSI     0x82
    MX8MQ_IOMUXC_ECSP1_MISO_ECSP1_MISO     0x82
    MX8MQ_IOMUXC_ECSP1_SS0_GPIO5_IO9       0x19
>;
};

pinctrl_spi2: spi2 {
    fsl,pins = <
        MX8MQ_IOMUXC_ECSP2_SCLK_ECSP2_SCLK    0x82
        MX8MQ_IOMUXC_ECSP2_MOSI_ECSP2_MOSI     0x82
        MX8MQ_IOMUXC_ECSP2_MISO_ECSP2_MISO     0x82
        MX8MQ_IOMUXC_ECSP2_SS0_GPIO5_IO13      0x19
    >;
};

```

&iomux 中引脚配置由 Config Tools for i.MX 工具自动生成或芯片手册(2198)。